Experiment-10:

Write a Python Program to Implement Expectation & Maximization Algorithm.

Code:

```
# ========================================
# Expectation-Maximization (EM) Algorithm
# Gaussian Mixture Model
# ========================================
from google.colab import files
uploaded = files.upload()
import pandas as pd
import io
# Read uploaded CSV
filename = list(uploaded.keys())[0]
df = pd.read_csv(io.BytesIO(uploaded[filename]))
print("First 5 Rows:")
print(df.head())
# Select Features
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]
# Feature Scaling
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
# Gaussian Mixture Model (EM Algorithm)
from sklearn.mixture import GaussianMixture
gmm = GaussianMixture(
    n_components=5,
    covariance_type='full',
    random_state=42
)
gmm.fit(X_scaled)

# Predict Cluster Labels
clusters = gmm.predict(X_scaled)
# Add Cluster Column
df['Cluster'] = clusters
print("\nCluster Assignment:")
print(df[['CustomerID',
          'Annual Income (k$)',
          'Spending Score (1-100)',
          'Cluster']].head(20))
# Cluster Counts
print("\nNumber of Customers in Each Cluster:")
print(df['Cluster'].value_counts().sort_index())
```

```
# Model Information
print("\nMeans of Gaussian Components:")
print(gmm.means_)
print("\nWeights of Gaussian Components:")
print(gmm.weights_)
print("\nLog Likelihood:")
print(gmm.score(X_scaled))
```

Output:

```
Mall_Customers[1].csv(text/csv) - 3981 bytes, last modified: 8/4/2026 - 100% done
Saving Mall_Customers[1].csv to Mall_Customers[1].csv
First 5 Rows:
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40

Cluster Assignment:
     CustomerID  Annual Income (k$)  Spending Score (1-100)  Cluster
0             1                  15                      39        4
1             2                  15                      81        2
2             3                  16                       6        4
3             4                  16                      77        2
4             5                  17                      40        4
5             6                  17                      76        2
6             7                  18                       6        4
7             8                  18                      94        2
8             9                  19                       3        4
9            10                  19                      72        2
10           11                  19                      14        4
11           12                  19                      99        2
12           13                  20                      15        4
13           14                  20                      77        2
14           15                  20                      13        4
15           16                  20                      79        2
16           17                  21                      35        4
17           18                  21                      66        2
18           19                  23                      29        4
```

```
18          19                   23                        29         4
19          20                   23                        98         2

Number of Customers in Each Cluster:
Cluster
0    84
1    39
2    21
3    33
4    23
Name: count, dtype: int64

Means of Gaussian Components:
[[-0.18510822 -0.030786  ]
 [ 0.98021511  1.23531662]
 [-1.35282052  1.16185843]
 [ 1.07840119 -1.3228595 ]
 [-1.27977698 -1.0882964 ]]

Weights of Gaussian Components:
[0.41333269 0.19669495 0.1035382  0.16552963 0.12090453]

Log Likelihood:
-2.262502621973241
```